



Introduction

•Rhythmic structure is central to AI-driven music and embodied choreography.

•In South Asian classical dance, rhythm is articulated through Nattuvangam, a **vocal–percussive system** of Solkattu (spoken syllables) and Talam (cyclic rhythmic structure).

•South Asian rhythmic systems are **underrepresented** in computational work. Limited digitization of Nattuvangam restricts computational study; **we introduce a curated low-resource dataset**.

What is Nattuvangam?

Nattuvangam is the vocal rhythmic art of South Asian classical dance, where **spoken syllables** (*solkattu*) are performed to articulate **cyclic rhythmic frameworks** (*tālam*).

Through *solkattu*, the performer explicitly encodes:

- **beat positions**
- **cycle boundaries**
- **tempo and rhythmic emphasis**

Spoken rhythm in Nattuvangam

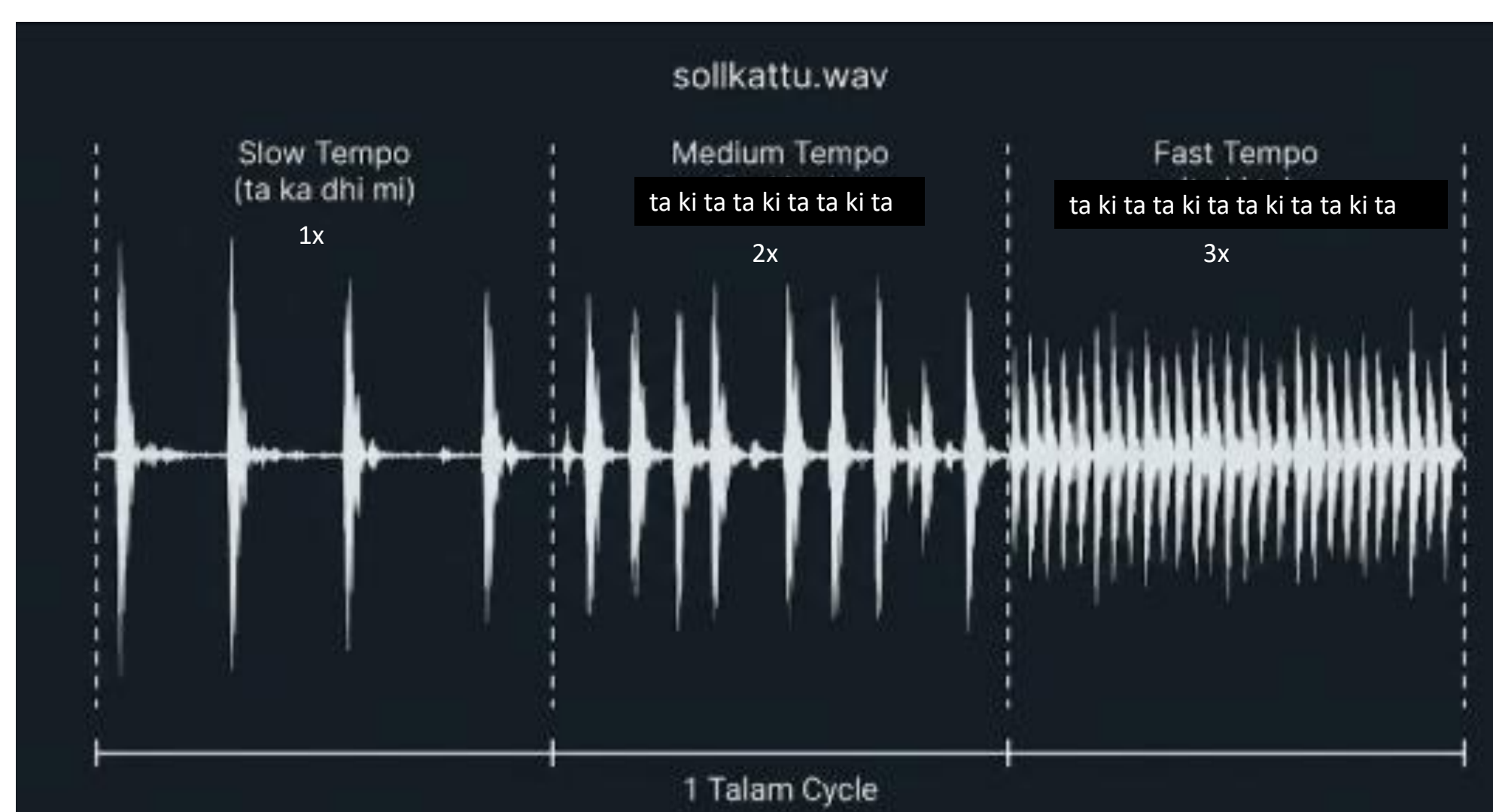
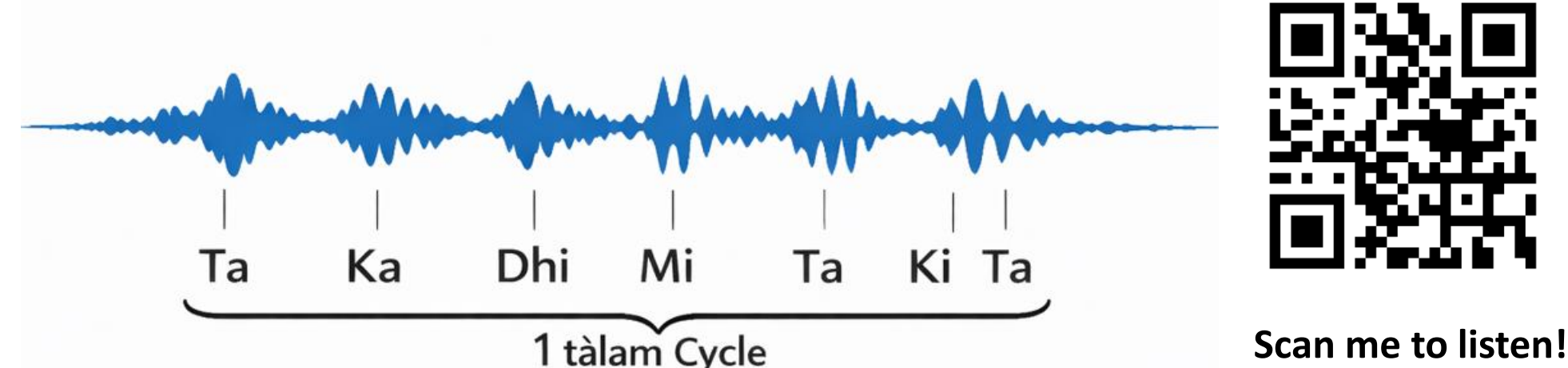


Fig 1: Example Nattuvangam solkattu rendered across tempos, illustrating explicit beat and cycle structure in low-resource audio.

Why does it matter?

- **Severely low-resource setting:** Nattuvangam recordings are scarce, short, and sparsely annotated, unlike the large, curated datasets used in most rhythm-learning work.
- **Structure is spoken, not inferred:** Rhythmic syllables (*solkattu*) explicitly encode beats and cyclic structure (*tālam*), reducing ambiguity in rhythm perception.
- **Direct link to movement:** Spoken rhythm is tightly synchronized with embodied dance actions, grounding rhythm in physical motion.
- **Why we study this:** Nattuvangam offers a rare testbed to evaluate whether machine learning models can learn beat and cycle structure from very limited audio when rhythmic structure is explicitly present.

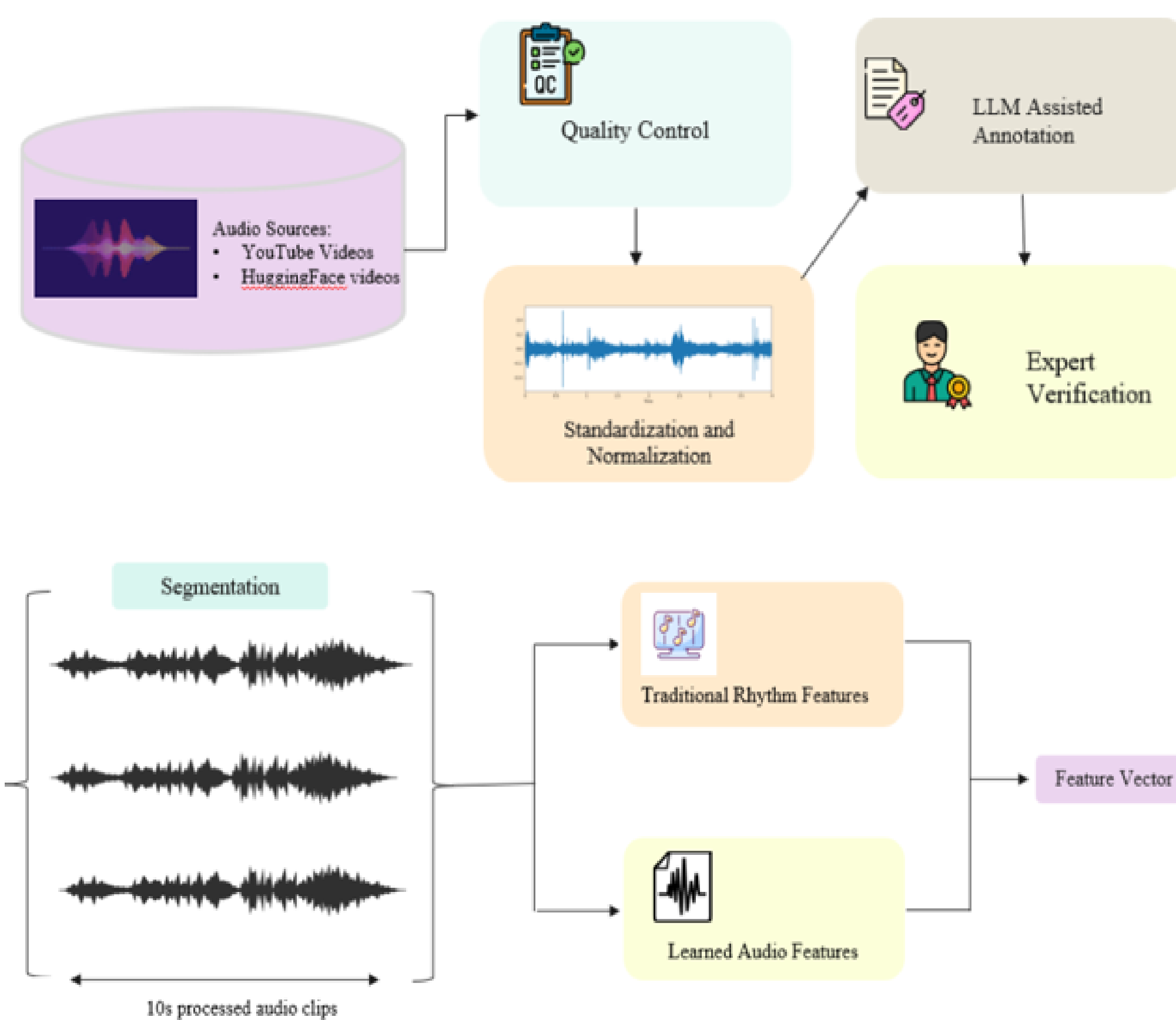


Dataset Creation & Overview (Low-Resource by Design)

We curate a **low-resource Nattuvangam dataset of 31560s (8.76 hours)** to study rhythmic structure under realistic data constraints.

Key characteristics:

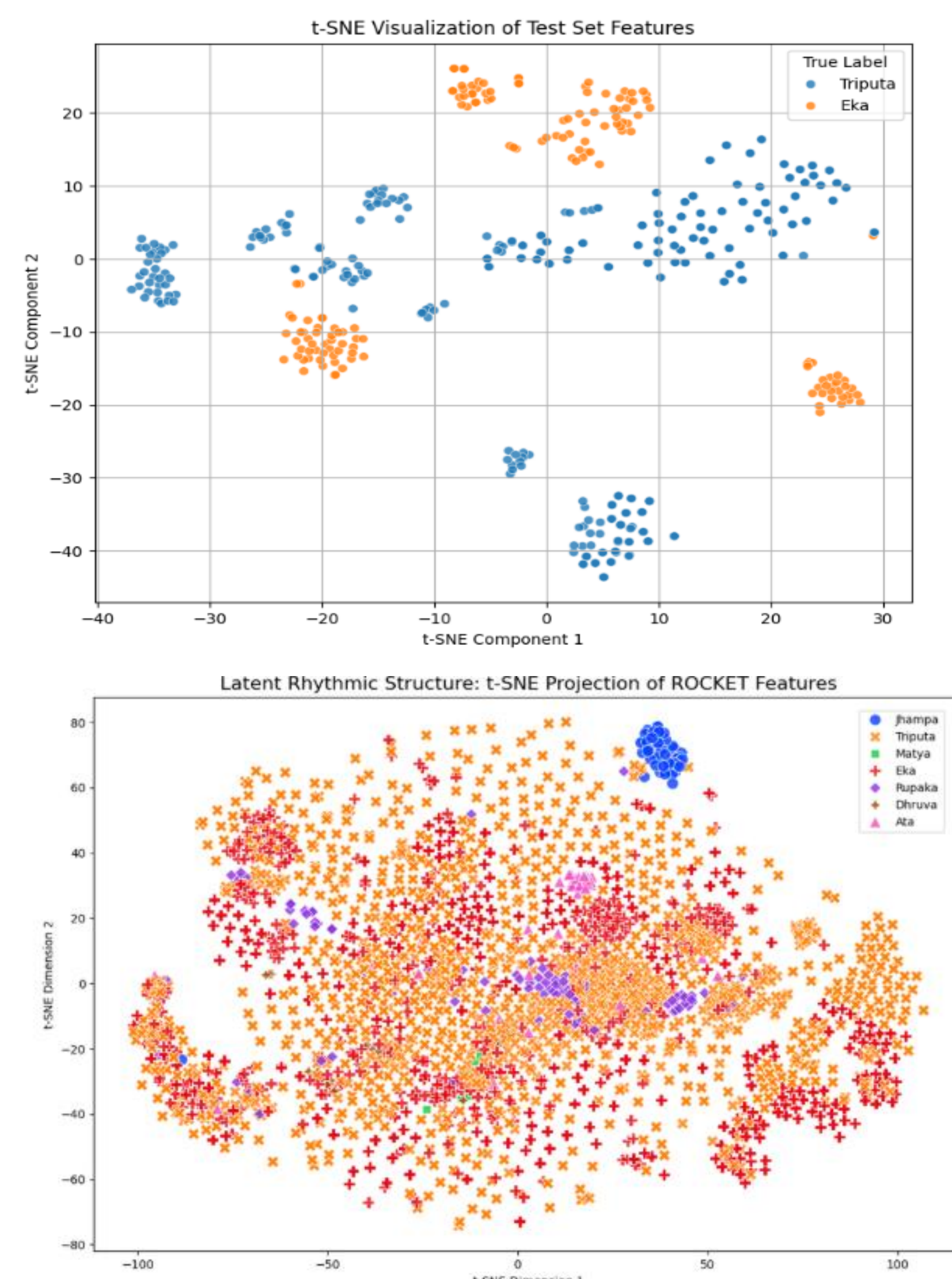
- **Short, standardized 10s audio clips** (spoken rhythm only)
- **Structured annotations:** *tālam* type(s), tempo (BPM), beat units (*akṣharas*), subdivisions (*aṅgas*), and spoken syllable sequences (*solkattu*), first solkattu, boolean_multiple_talas
- **Scalable annotation pipeline:** LLM-assisted labeling with expert verification



Feature Generalization in Low-Resource Audio

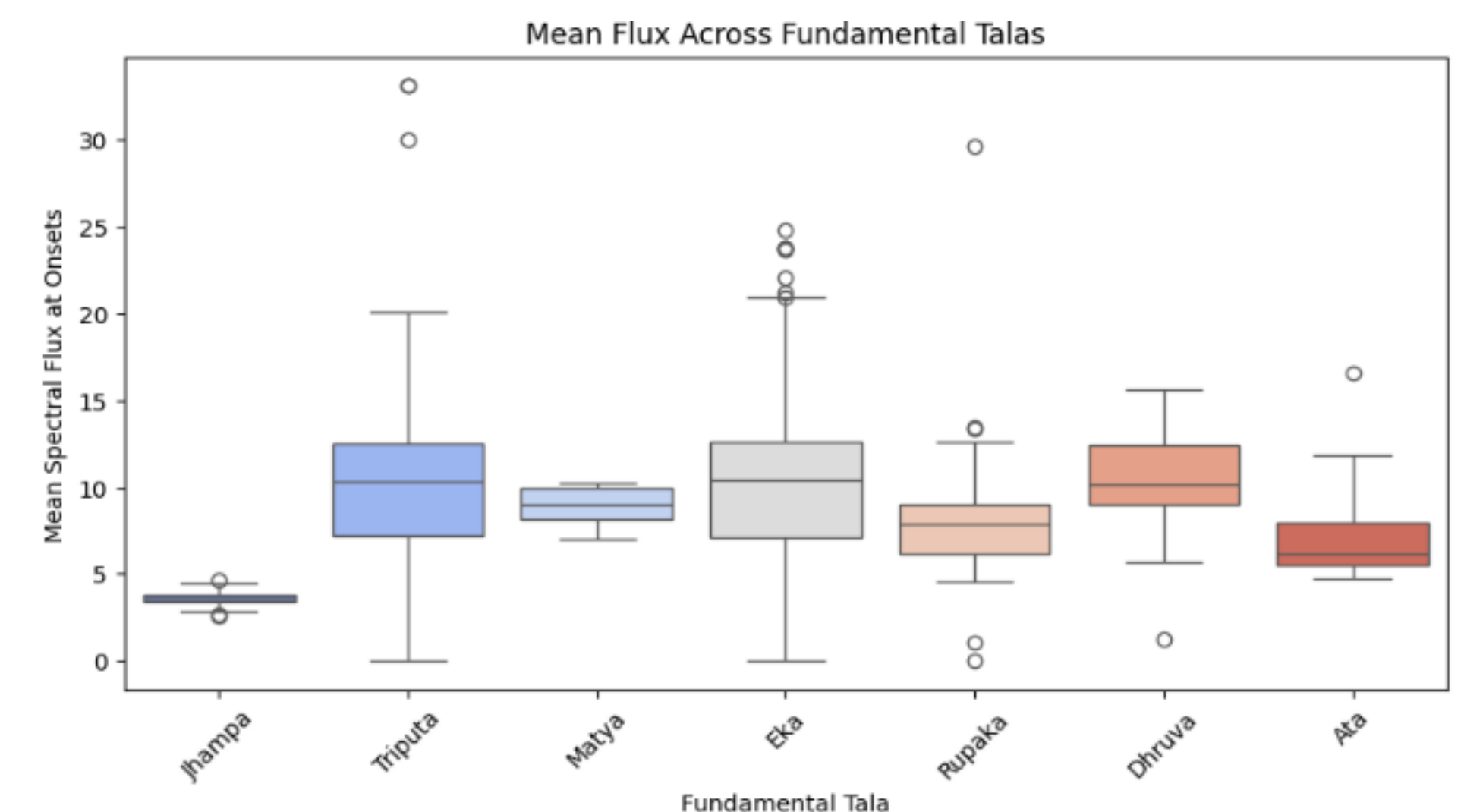
We compare a simple audio baseline (ROCKET + MFCC) with hybrid features that include pretrained CLAP embeddings for tāla recognition.

Clearer t-SNE clusters with CLAP show that pretrained models capture transferable rhythmic patterns, even from very limited audio data.

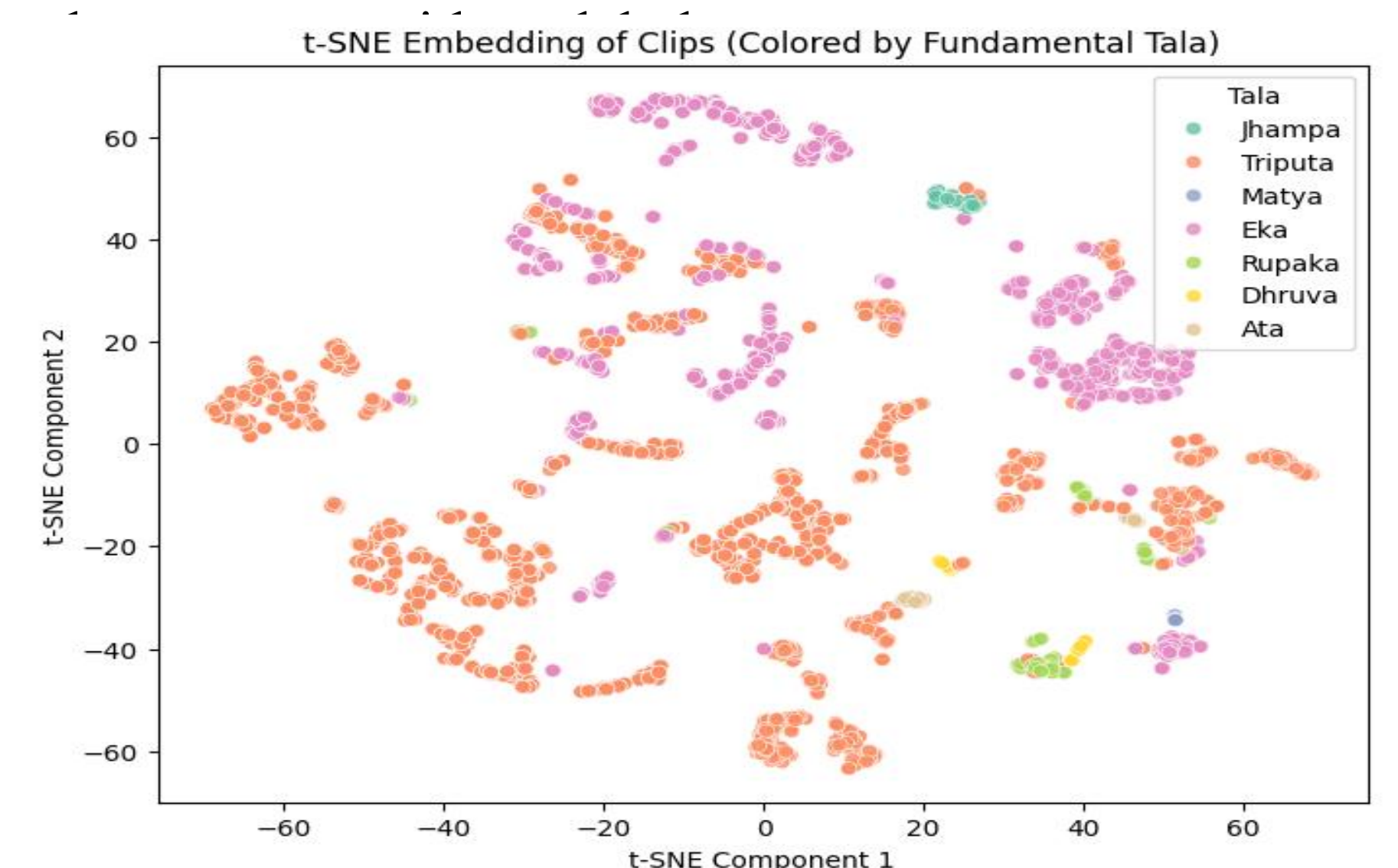


Observations

Rhythmic traits & Solkattu: Tempo, spectral flux, and IOIs differ across tālas, showing expressive vs. stable rhythms. Full and partial first solkattu cycles guide alignment, boundary variability, and transfer learning for rare tālas.



Modeling insights: Models trained on common tālas capture general timing and accents transferable to rare tālas. MFCC + CLAP embeddings improve recognition, and PCA/t-SNE reveal natural tāla



Broader Relevance and Future Work

- Provides a general framework for learning structure from low-resource spoken or percussive audio across languages and cultures.
- Applicable to other low-resource settings where rhythm, prosody, or timing is explicit, but data is scarce.
- Complements our parallel work on fine-grained 3D motion modeling of expressive South Asian dance using human mesh representations.
- Future work includes joint audio–motion learning, cyclic boundary modeling, and zero-shot recognition of unseen rhythmic forms.

Conclusion

•Nattuvangam makes rhythmic structure explicit through spoken syllables and tāla cycles.

•Despite extreme data scarcity, the dataset preserves clear, learnable rhythmic signatures.

•This work demonstrates a viable path for low-resource, culturally grounded rhythm learning.

References

1. D. Anantha Rao. Tala and its significance. Naad – Nartan Journal of Dance and Music, 11:5–8, May 2023. ISSN 2349-4654.
2. Yanjie Sun, Kele Xu, Chaorun Liu, Yong Dou, Huaimin Wang, Bo Ding, and Qinghua Pan. Automated data augmentation for audio classification. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 32:2716–2728, 2024.
3. Wangduk Seo, Sung-Hyun Cho, Pawel Teisseyre, and Jaesung Lee. A short survey and comparison of cnn-based music genre classification using multiple spectral features. IEEE Access, 12:245–257, 2023.

